Classroom Activity - 2

CSA 1614 - Data Warehousing and Data Mining (DWDM)

Name: _____ Register No: _____ Date: 12-10-2026

Task 1: Create a Sales Data Warehouse

Unit I - Data Warehousing | *Time: 30 minutes*

1.1 Aim

To create a small sales dataset of 10 records in Excel, identify the fact table and dimension tables, draw a star schema, and write two questions that the data warehouse can answer.

1.2 Sales Dataset (10 records)

The dataset has five columns: Product, City, Month, Quantity and Sales Amount (in Rupees).

S.No	Product	City	Month	Quantity	Sales Amount (Rs.)
1	Laptop	Chennai	January	5	2,50,000
2	Mobile	Chennai	January	10	1,50,000
3	Laptop	Mumbai	February	3	1,50,000
4	Tablet	Delhi	February	7	1,40,000
5	Mobile	Mumbai	March	12	1,80,000
6	Tablet	Chennai	March	4	80,000
7	Laptop	Delhi	April	6	3,00,000
8	Mobile	Delhi	April	8	1,20,000
9	Tablet	Mumbai	May	5	1,00,000
10	Laptop	Chennai	May	4	2,00,000

Total Quantity = 64 units. Total Sales Amount = Rs. 16,70,000.

1.3 Fact Table and Dimension Tables

A data warehouse stores data in a multidimensional form. Numeric values that are measured and added up are stored in the fact table. Descriptive labels used to group, filter and analyse those numbers are stored in dimension tables.

Fact Table: Sales_Fact

Sales_Fact stores the measures Quantity and Sales_Amount, along with foreign keys that link to each dimension table.

Sale_ID	Product_ID (FK)	City_ID (FK)	Time_ID (FK)	Quantity	Sales_Amount
1	P1	C1	T1	5	2,50,000
2	P2	C1	T1	10	1,50,000
3	P1	C2	T2	3	1,50,000
4	P3	C3	T2	7	1,40,000
5	P2	C2	T3	12	1,80,000
6	P3	C1	T3	4	80,000
7	P1	C3	T4	6	3,00,000
8	P2	C3	T4	8	1,20,000
9	P3	C2	T5	5	1,00,000
10	P1	C1	T5	4	2,00,000

Dimension Tables

Product_Dim

Product_ID	Product_Name
P1	Laptop
P2	Mobile
P3	Tablet

City_Dim

City_ID	City_Name
C1	Chennai
C2	Mumbai
C3	Delhi

Time_Dim

Time_ID	Month
T1	January
T2	February
T3	March
T4	April
T5	May

Summary

- **Fact table:** Sales_Fact (Quantity, Sales_Amount).
- **Dimension tables:** Product_Dim, City_Dim, Time_Dim.
- Tip to remember: numbers you add up are facts; labels you group by are dimensions.

1.4 Star Schema

In a star schema, the fact table sits in the centre and each dimension table is connected directly to it. The diagram looks like a star, which gives the schema its name. PK means primary key and FK means foreign key.

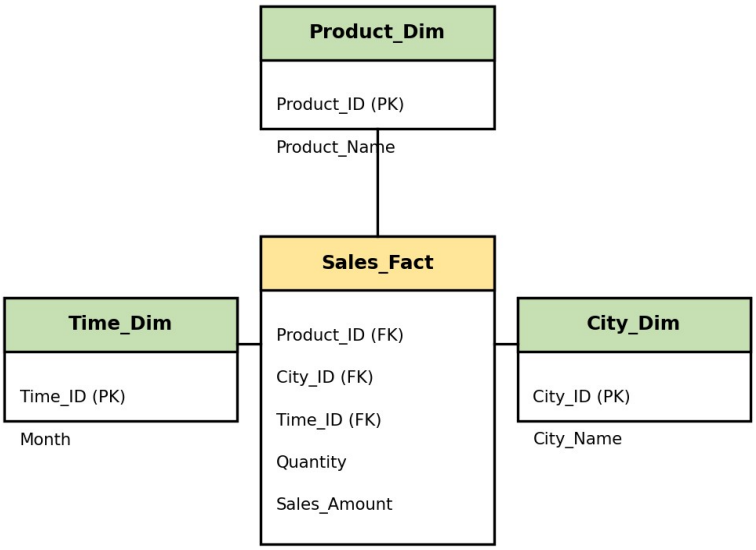


Figure 1: Star schema for the sales data warehouse

Features of this schema

- Sales_Fact is the central table and contains one row per sale.
- Each dimension has a primary key (Product_ID, City_ID, Time_ID) that appears as a foreign key in the fact table.

1.5 Questions the Warehouse Can Answer

1. **What is the total sales amount of each product in each city?**

Answer from the data:

Product	Chennai (Rs.)	Mumbai (Rs.)	Delhi (Rs.)
Laptop	4,50,000	1,50,000	3,00,000
Mobile	1,50,000	1,80,000	1,20,000
Tablet	80,000	1,00,000	1,40,000

Laptop in Chennai gives the highest amount (Rs. 4,50,000). Overall, Laptop earns Rs. 9,00,000, Mobile Rs. 4,50,000 and Tablet Rs. 3,20,000.

2. **Which month had the highest total sales quantity?**

Month	Total Quantity
January	15
February	10
March	16
April	14
May	9

March had the highest total quantity with 16 units.

Task 2: Data Cleaning Challenge

Unit II - Data Preprocessing | *Time: 20 minutes*

2.1 Aim

To prepare a table of 10 customer records containing missing values, duplicate records and spelling errors, then identify the data quality problems, remove duplicates, handle missing values and submit the corrected dataset.

2.2 Messy Customer Dataset (before cleaning)

Cust_ID	Name	City	Age	Email
1	Ravi Kumar	Chenai	25	ravi@gmail.com
2	priya sharma	Mumbai		priya@gmail.com
3	Arun Raj	Delhi	30	arun@gmail.com
4	Arun Raj	Delhi	30	arun@gmail.com
5	Meena S	Banglore	28	
6	Karthik	Chennai	35	karthik@gmail.com
7	Divya R	Mumbay	22	divya@gmail.com
8	suresh M	Delhi		suresh@gmail.com
9	Anitha K	Chennai	27	anitha@gmail.com
10	Vijay P	Hyderabad	31	vijay@gmail.com

2.3 Data Quality Problems Identified

Problem	Where it occurs	Why it is a problem
Spelling errors	Chenai, Banglore, Mumbay	The same city appears under different names, so grouping gives wrong counts.
Duplicate record	Arun Raj (Cust_ID 3 and 4)	The customer is counted twice, which inflates totals.
Missing Age	Cust_ID 2 and 8	Age calculations and averages become inaccurate.
Missing Email	Cust_ID 5	The customer cannot be contacted by email.
Inconsistent capitalization	priya sharma, suresh M	Text values look inconsistent and may not match during searches.

2.4 Cleaning Steps in Excel

All cleaning was done on a copy of the sheet named Cleaned Data, so the original messy data stays unchanged.

1. **Remove duplicates:** Select the table, go to Data > Remove Duplicates, untick Cust_ID, keep the other columns ticked and click OK. The duplicate record of Arun Raj (Cust_ID 4) is removed and 9 records remain.
2. **Fix spelling errors:** Press Ctrl+H (Find and Replace). Replace Chenai with Chennai, Banglore with Bangalore and Mumbay with Mumbai, using Replace All each time.
3. **Fix capitalization:** In a helper column type =PROPER(B2) and fill down. Copy the result and use Paste Special > Values over the Name column, then delete the helper column. priya sharma becomes Priya Sharma and suresh M becomes Suresh M.
4. **Handle missing Age values:** Use =AVERAGE(D2:D10). The available ages (25, 30, 28, 35, 22, 27, 31) add up to 198, and 198 / 7 = 28.29, which is rounded to 28. This value is entered in the blank Age cells.
5. **Handle missing Email:** The blank email of Meena S is replaced with the text Unknown.
6. **Rename columns clearly:** Cust_ID becomes Customer_ID and Name becomes Customer_Name.

2.5 Cleaned Dataset (after cleaning)

Customer_ID	Customer_Name	City	Age	Email
1	Ravi Kumar	Chennai	25	ravi@gmail.com
2	Priya Sharma	Mumbai	28	priya@gmail.com
3	Arun Raj	Delhi	30	arun@gmail.com
5	Meena S	Bangalore	28	Unknown
6	Karthik	Chennai	35	karthik@gmail.com
7	Divya R	Mumbai	22	divya@gmail.com
8	Suresh M	Delhi	28	suresh@gmail.com
9	Anitha K	Chennai	27	anitha@gmail.com
10	Vijay P	Hyderabad	31	vijay@gmail.com

2.6 Result

- Original records: 10. Records after removing the duplicate: 9.
- Spelling errors corrected: 3. Capitalization errors corrected: 2.
- Missing values handled: 3 (two Age values filled with 28 and one Email set to Unknown).

Conclusion

A star schema with one fact table and three dimension tables was designed for the sales data, and it can answer questions such as total sales by product and city. In the second task, data quality problems were identified and corrected, giving a clean dataset that is ready for analysis.